

TUSHAR BANSAL

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Education

Indian Institute of Technology, Madras
B.S. (Data Science and Applications); CGPA: 7.62

2022 – Present
Chennai, Tamil Nadu

Rajasthan Technical University
B.Tech. (IT); CGPA: 9.01

2023 – Present
Jaipur, Rajasthan

Technical Skills

Languages & Modeling: Python, Java, SQL, R, Bash, C, C++
Frameworks & Libraries: PyTorch, Scikit-Learn, Hugging Face, NumPy, Pandas
Architecture & Web MLOps: Flask API, Vue.js, Redis, Celery Task Queues, Asynchronous Architecture
Developer Tools & DevOps: Git, Linux, CI/CD Pipelines, Docker, Sed, Awk, PostgreSQL, SQLite
Domain Expertise: Deep Learning, Machine Learning, Neural Networks, Distributed Systems, RDBMS

Experience

Molde University College, Norway
Research Intern (Supervisor: Dr. Swati Aggarwal, Professor of AI)

May 2026 – Present
Molde, Norway

- Investigating trustworthy AI methods for pediatric cardiac screening using phonocardiogram (PCG) signals, integrating self-supervised learning, uncertainty quantification, and explainable AI on the CirCor dataset.

Creware Technologies Pvt. Ltd.
Software Engineer Intern

June 2025 – August 2025
Jaipur, Rajasthan

- Architected a Rasa-based conversational **AI assistant using Rasa NLU and SDK**, optimizing intent classification across **1,000+ training variants per intent** and designing multi-turn booking workflows that validated user inputs, **automated backend API transactions**, and triggered operational email notifications.

Indian Institute of Technology, Madras
Teaching Assistant for DBMS

May 2023 – May 2024
Chennai, Tamil Nadu

- Supported instruction for **300+ students in IIT Madras' DBMS course**, guiding learners through database design (RDBMS, normalization), indexing (Hash tables, B/B+ Trees), and transaction management (ACID compliance), while building practical PostgreSQL exercises for hands-on learning.

Publications & Research Papers

T. Bansal et al., "Clustering Districts by Educational Performance Using Machine Learning and Neutrosophic Logic", Presented at Springer International Conference on Emerging Trends in Engineering and Smart Systems (ICETESS), Jaipur, India, 2026. | [Presented Paper](#)

Projects

Music Genre Classification under Distribution Shift | *PyTorch, Transformers* | [GitHub](#) **April 2026**

- Mitigated distribution shift in music genre classification by engineering a data augmentation pipeline with **5,000+ synthetic audio mashups** and ESC-50 environmental noise, then benchmarking CNN, CNN-BiLSTM, and **86M-parameter Audio Spectrogram Transformer** models to achieve a peak **0.89555 F1-score** on Kaggle using mixed-precision training, test-time augmentation, and label smoothing.

Comment Category Classification with Massive Class Imbalance | *LightGBM, Scikit-Learn* | [GitHub](#) **April 2026**

- Developed a text moderation system by engineering **65,000+ lexical, sentiment, profanity, and n-gram features** from user comments and training an optimized **LightGBM ensemble**, achieving a **0.82 Macro F1-score** despite a **20x class imbalance** between severe/violent and benign content.

Distributed Hospital Management System | *Flask, Vue.js, Redis, Celery* | [GitHub](#)

December 2025

- Architected a full-stack hospital management platform featuring secure authentication, role-based access control, and Redis-Celery driven asynchronous workflows, enabling automated reporting, webhook integrations, and high-performance cached APIs.

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | [Certificate Link](#)

October 2025

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | [Certificate Link](#)

October 2025

Oracle AI Vector Search Certified Professional | [Certificate Link](#)

October 2025